



2023 NASA Student Launch – Proposal

UNT Robotics: Aerospace Division

University of North Texas: College of Engineering

3940 N Elm St, Denton TX 76207



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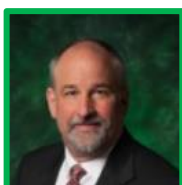


3. Team Information and Organizational Structure

3.1. Objective

The University of North Texas intends to compete in the 2022-23 NASA Student Launch competition. The competition provides opportunities to develop new engineering, communication and soft skills and apply knowledge learnt in the classroom to a real project. Competing gives a platform for students to solve multi-discipline advanced engineering problems in a practical setting. We hope the challenges and constraints provided throughout the competition will better set up students for future employment. Furthermore, we intend to bring opportunities to the university through sponsor, industry and faculty engagement and the team STEM outreach focus brings many opportunities and new connections in the local community. Our overall object is the development and improvement of the students, team, university, and community.

3.2. Educator Information



Dr. Mark Wasikowski

Title: Faculty Advisor/Mentor

Phone: 940-369-8030

Email: mark.wasikowski@unt.edu



Dr. Hassan Qandil

Title: Faculty Advisor/Mentor

Phone: 940-565-4205

Email: Hassan.Qandil@unt.edu

3.3. NAR Mentor



Jack Sprague (HPR LVL 3)

TRA #: 1467

NAR #: 45689

Phone: 940-321-2132

Email: jsprague@ix.netcom.com

Suzy Sprague (Rocket Mom)



3.4. Student Team Leader



Joseph Moore (HPR LVL 1)

Aerospace Division Lead/Project Manager

NAR #: 110792

TRA#: 23072

Senior Mechanical and Energy Eng. B.S.

Phone: 760-583-7155

Email: josephmoore2@my.unt.edu

3.5. Safety Officer



Benjamin Bailey

Junior Mechanical and Energy Eng. B.S.

Phone: (254) 534-0828

Email: BenjaminBailey@my.unt.edu

3.6. Student Design Leaders

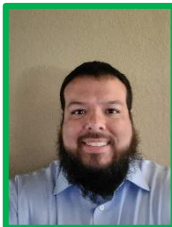


Keon Brown

Senior Mechanical and Energy Eng. B.S.

Phone: 469-386-6537

Email: keonbrown@my.unt.edu



Tommie Snow

Senior Mechanical and Energy Eng. B.S.

Phone: 214-918-9631

Email: TommieSnow@my.unt.edu





Brian Garcia

Senior Mechanical and Energy Eng. B.S.

Phone: 214-962-7100

Email: briangarcia@my.unt.edu



Garrett Kropp

Senior Mechanical and Energy Eng. B.S.

Phone: 916-899-4096

Email: garrettkropp@my.unt.edu



David Nachega

Senior Mechanical and Energy Eng. B.S.

Phone: 469-478-6589

Email: davidnachega@my.unt.edu



3.7. Organizational Structure

There are 14 students dedicated to this project, six of whom are senior mechanical engineering students working Project Phoenix as their capstone (senior design) project. Everyone involved will be participating in multiple sections of the project, but there are individuals responsible for major work areas to ensure continuity for reporting and meeting the planned production cycle milestones. These are some of the key areas that we have identified as essential for the planning, production, and development process:

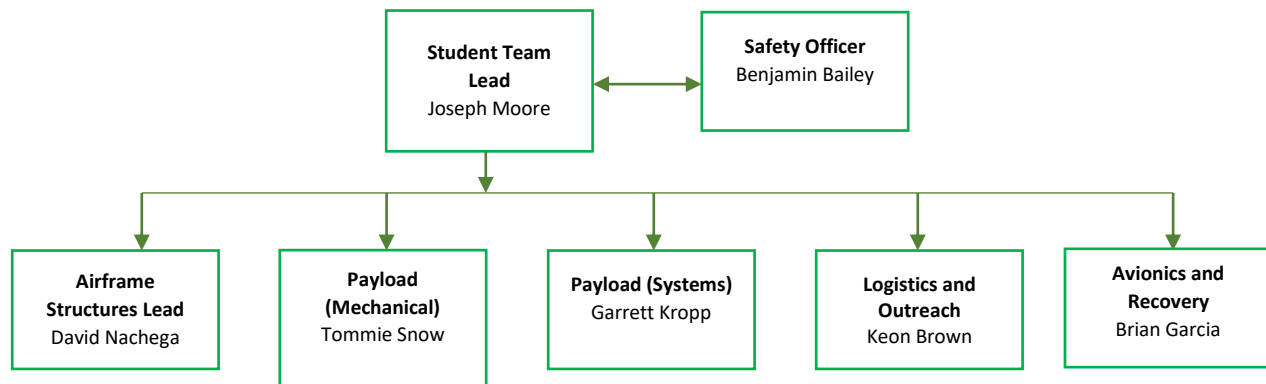


Figure 1: Team Organizational Flow Chart

3.8. NAR/TRA Sections

The team has access to multiple launch sites in our area offering at least two launch opportunities per month. Tripoli North Texas (TNT Prefecture # 58) and the Dallas Area Rocketry Society (aka DARS, NAR section 308) each have multiple launch sites within two hours of the university that we intend to utilize for our sub-scale and full-scale rocket launches. The team will also be receiving mentoring from Jack Sprague, NAR and TRA LVL 3 certified, and a member of both TNT and DARS. Several members of our team are either TNT or DARS members, which will help facilitate planning and communication through these groups.

3.9. Hours Invested in Proposal

Through brainstorming, attending meetings, researching, and writing the proposal, the team has invested 115 hours to date.

4. Facilities and Equipment

We have several facilities and labs available to the team during the build process, including a dedicated workshop, university manufacturing lab and a senior design lab. Furthermore, we have plenty of materials and equipment available to ensure we can build the necessary vehicles and systems.



4.1. University of North Texas Discovery Park: Aerospace Division Workspace

Description

Our team has a dedicated workspace in the manufacturing area of Discovery Park. Full-scale launch vehicle manufacturing, planning, fabrication, and assembly is conducted in this area. Any energetic testing of subassemblies will be conducted in the open spaces surrounding Discovery Park to avoid injury to personnel or damage to property.

Hours and Required Personnel

This facility is available eight hours a day, five days a week. Special requests can be made to accommodate after hours and weekend hours. A faculty member or UNT staff will need to be present to conduct work in the project area.



Figure 2: Aerospace Division Workspace

4.2. University of North Texas Discovery Park: Engineering Manufacturing Facility (EMF) Lab

Description

Team members will have access, once proper safety training is conducted, to the EMF Lab at Discovery Park. The EMF Lab is dedicated to offering advanced manufacturing services such as fabrication of mechanical prototypes, proof of concepts for machinery, 3D Printing, Laser Engraving, and CNC precision machining. This will allow production of machined airframe or payload parts in house and in a controlled environment.



Figure 3: EFM Lab

Hours and Required Personnel

This facility is available Monday through Friday from 8am to 4:30pm. There is always a qualified person employed by the university present during these hours. For our own safety precautions, the rocketry team members will only operate these machines in teams of two people, one of whom will be a qualified safety officer.



4.3. Discovery Park Senior Design Lab

Description

The Senior Design lab is a dedicated workspace for the current seniors to work on their capstone designs. We will use this lab to do any composites work during the manufacturing process because the area is climate-controlled and large enough to allow us to roll a large composite airframe tube, if needed. Subscale vehicle and payload design, prototyping and testing, manufacture and assembly will also be conducted in this lab.

Hours and Required Personnel

The lab is available during working hours, five days a week, and 10am to 7pm on Saturdays.

4.4. Required Supplies

- | | |
|---|---|
| <ul style="list-style-type: none">• Masking Tape• Mechanics gloves• Latex/vinyl gloves• Black Powder• Electric Matches• Fireproof Insulation• Epoxy Resin• Fiberglass/Carbon fiber fabric• Paint rollers and plastic spatulas• Mylar rolls | <ul style="list-style-type: none">• Balsa Plywood• Wood Glue• Cyanoacrylate (CA)• Various 3D printing filaments• Nylon shear pins and rivets• Various electronic components• Altimeters, batteries, etc.• Peel ply• Respirators |
|---|---|

4.5. Manufacturing Equipment

- | | |
|---|---|
| <ul style="list-style-type: none">• Lathe• Laser cutting machine• Various 3D Printing equipment• Hand tools• Table Saw• Router• Drill | <ul style="list-style-type: none">• CNC Machine• Water Jet machine• Rolling Setup for Mandrel• Vacuum pump system for vacuum impregnation• Various molds for fiberglass/carbon fiber production |
|---|---|

4.6. Testing Supplies

- | | |
|---|---|
| <ul style="list-style-type: none">• Safety Glasses• Precision Electronic Scale• Remote Igniter system• High Speed Camera• Timer/Stopwatch | <ul style="list-style-type: none">• Voltmeter• Thermal Camera• Material Science Lab (for testing mechanical properties of material samples) |
|---|---|



4.7. Computer Software

- SolidWorks: CAD design and FEA analysis
- MATLAB: Control systems design
- ANSYS: CFD analysis
- Discord: Virtual meeting and collaboration
- Google Docs: Document sharing, storage, and editing
- OpenRocket: Launch vehicle design, configuration planning, and simulation
- Visual Studio: writing and editing software to be used in any subsystems and payload

5. Safety

The high-power rocketry team at the University of North Texas holds the safety and well-being of its members, supporters, and observers as the highest priority. Overall responsibility for safety falls to the Safety Officer and Project Manager. Additionally, each member of the team is tasked with having the skills and knowledge to identify potential hazards and to use the appropriate safety procedures to mitigate them. To ensure all team members involved in the production, testing, and operational process are aware of the hazards associated with general construction and the hazards inherent to high power rocketry, all team members will be required to read the team safety SOP, enclosure (1), and sign it acknowledging that they understand the hazards and will follow procedures to mitigate the risks.

5.1. General Safety Requirements [1]

1. Teams must utilize all launch and safety checklists.
2. The team must identify a student safety officer who is responsible for all items in section 5.3 of the NASA SL Handbook.
3. Teams must abide by all rules and regulations of the RSO for local launches for all test flights.
4. Teams must abide by all FAA rules and regulations.
5. Teams must abide by all FCC rules and regulations.
6. All team members understand and will abide by the following safety regulations:
 - a. Range safety inspections will be conducted on each rocket before it is flown. Each team shall comply with the determination of the safety inspection or may be removed from the program.
 - b. The Range Safety Officer has the final say on all rocket safety issues; therefore, the Range Safety Officer has the right to deny the launch of any rocket for safety reasons.
 - c. The team mentor is ultimately responsible for the safe flight and recovery of the team's rocket. Therefore, a team will not fly a rocket until the mentor has reviewed the design, examined the build, and is satisfied the rocket meets established amateur rocketry design and safety guidelines.
 - d. Any team that does not comply with the safety requirements will not be allowed to launch their rocket.



5.2. Safety Plan

The team must maintain a safety plan throughout the project to ensure safety measures are properly followed:

1. Responsibilities will be assigned to an officer for oversight of our safety procedures.
2. The team will identify hazards and characterize all mitigation strategies.
3. The team will create detailed procedures for all testing, manufacturing, and launches.
4. Extensive testing of components will be done to ensure safety and reliability.

5.3. Risk Assessment and Mitigation Factors

Our goal through applying risk mitigation and making risk decisions is to lower the risk assessment code (RAC) for each activity by at least one RAC value.

Risk Severity		Risk Probability			
		Likely/Immediate A	Probably B	May C	Unlikely D
Catastrophic	1	1A	1B	2C	3D
Critical	2	1A	2B	3C	4D
Moderate	3	2A	3B	3C	5D
Minor	4	3A	4B	5C	5D
Negligible	5				

Table 1: Risk Assessment Matrix [2]

5.4. Risk Assessment Worksheet

Operational Risk Management Assessment Worksheet:							
Identify Hazards			Assess Hazards	Make Risk Decisions		Implement Controls	Supervise
Operation	Hazard	Causes	Initial RAC	Develop Controls	Residual RAC	How to Implement	How to Supervise
Fiberglass Fabrication	Epoxy resin skin contact, eye contact, fumes inhalation	Chemical spills, burns from curing epoxy, and reaction fumes	2A	Eye pro, gloves, respirators, and protective clothing will be worn during operations	4B	PPE will be available in the workspace, and all team members (TM's) will be trained and briefed on safety hazards present.	There will be a Safety officer or designated person present for all hazardous operations.
General Parts Construction	Serious injury	Power tools malfunction, inherent hazards associated with, or improper use of tools.	2C	Wearing eye pro, ear pro, and work gloves while using appropriate machinery. Train personnel on proper use and troubleshooting of tools.	3D	PPE will be available in the workspace. All personnel admitted to certain areas will finish required hazard training. TM's will be	There will be a Safety officer or designated person present for all hazardous operations.



						trained and briefed on hazards present.	
Energetic Testing of components and subsystems	Burn or impact injury	Black powder ejection charges	2C	Wearing eye pro, ear pro, and work gloves while around testing area. Train personnel on proper use and troubleshooting of component systems.	3D	PPE will be available in the workspace. All personnel admitted to certain areas will finish required hazard training. TM's will be trained and briefed on hazards present.	There will be a Safety officer or designated person present for all hazardous operations.
	Shock/electrocution	Electrical circuit testing	3C	Ensure proper grounding and handling techniques area followed.	5D	PPE will be available in the workspace. All personnel admitted to certain areas will finish required hazard training. TM's will be trained and briefed on hazards present.	There will be a Safety officer or designated person present for all hazardous operations.
	Impact injury	Actuator malfunction	3C	Wearing eye pro and gloves while working with any energetic systems.	5D	PPE will be available in the workspace. All personnel admitted to certain areas will finish required hazard training. TM's will be trained and briefed on hazards present.	There will be a Safety officer or designated person present for all hazardous operations.
Painting Launch Vehicle Components	Skin/eye contact, fumes	Spray Paint and clear coat	2A	Eye pro, gloves, particulate masks, and protective clothing will be worn during operations. Masks can be omitted if painting is done outside.	4B	Workspace is in a well-ventilated area. PPE will be available to all TMs in the workspace. TM's will be trained and briefed on hazards present.	There will be a Safety officer or designated person present for all hazardous operations.
Launch Day and Test Launches	Loss of vehicle	Motor Failure	2C	Only use COTS motors.	3D	Inspect motor upon receipt and before use.	Records of this inspection will be kept by the project manager.
	Range Fire	Motor failure, motor retainer	2C	Only use COTS motors. Follow NAR/Tripoli range	3D	Inspect motor upon receipt and before use.	The project manager must be present and



		failure		safety guidelines for range fires. Conduct modeling and testing of motor retention system.		Inspect retention system when assembled and as part of the pre-flight checklist at each launch.	inspect this component before each launch.
	Graphite powder inhalation or eye contact	Dry lubricant aerosolization during application to mechanical parts.	3C	Use gloves, eye protection, and masks/respirators when applied indoors. Use point application bottle directly at the application point to minimize wind effects.	5D	PPE will be available in the workspace. All personnel admitted to certain areas will finish required hazard training. TM's will be trained and briefed on hazards present.	
	Injury to personnel or damage to property, loss of vehicle or components	Detonation/ Deflagration of Motor	2C	Only use COTS motors. Follow NAR/Tripoli range safety guidelines for minimum safe distances. Follow NASA SL guidelines for vehicle construction to minimize the likelihood of projectile/ fragmentation production.	3D	Inspect motor upon receipt and before use. Utilize NAR/Tripoli minimum safe distances and follow instructions from the RSO and RCO of the launch.	Project manager, or designated safety person will inspect components during and after construction. Components will be inspected before and after each launch to ensure they are safe to reuse or identify the need for replacement.
	Loss of vehicle/ damage to property/ personal injury	Recovery system failure	1B	Charge all batteries, bring extras to each launch. Inspect parachute and drogue, and all lines. Verify ejection charges by ground testing prior to launches. Follow NAR guidelines for weathercocking to prevent vehicle from falling on top of people/property at the range. Build in redundancies to ensure that there are backup recovery systems.	2C	Follow pre-flight checklists to ensure that each system is ready and working properly.	Project Manager and safety officer will sign off on each pre-flight checklist prior to declaring a vehicle ready to launch. If any part of the checklist is incomplete the vehicle launch will be postponed until launch readiness can be verified.
	Fin Damage	Recovery system Failure or landing speed miscalculation	2C	Build for redundancy in the recovery system. Reinforce fins with carbon fiber to increase shear strength and prevent breaking. Design fin-can to be swapped	3D	Follow pre-flight checklists to ensure that each system is ready and working properly.	Project Manager and safety officer will sign off on each pre-flight checklist prior to declaring a vehicle ready to launch. If any part of the



				out if necessary. Conduct failure testing to determine strength of fins before assembly.			checklist is incomplete the vehicle launch will be postponed until launch readiness can be verified.
	Body Tube Failure	Zippering caused by deployment of recovery system	2C	Base material strength on worst case. Employ “zipper-proof” design techniques wherever possible in construction.	3D	Stress-test all airframe elements to ensure survivability during in-flight events. Redesign or reinforce when necessary.	Before assembly of flight ready vehicle, production checklist must be signed by project manager and manufacturing lead to ensure that each element meets the stress even criteria established during design phase and testing.
	Body Tube Failure	Forces exerted at launch, during flight, recovery events, or landing exceed design parameters	2C	Base material strength on worst case. Limit weight of launch vehicle only when it does not sacrifice required strength. Conduct FEA and CFD/wind tunnel testing to predict worst case forces on airframe components.	3D	Stress-test all airframe elements to ensure survivability during in-flight events. Redesign or reinforce when necessary.	Before assembly of flight ready vehicle, production checklist must be signed by project manager and manufacturing lead to ensure that each element meets the stress even criteria established during design phase and testing. All elements will undergo necessary ground testing prior to certification for flight.
	Premature Separation of Airframe Sections	Expansion of trapped gasses within sealed sections, forcing stages apart	3C	All airframe components that would normally be sealed will have appropriately sized vent holes added to allow trapped gasses to escape as they expand.	4D	Once all components are ready to be assembled, they will be inspected to ensure that the correct holes have been added prior to prepping for launch.	Before assembly of flight ready vehicle, production checklist must be signed by project manager and manufacturing lead to ensure that each element meets the stress even criteria established during design phase and testing. All elements will undergo necessary ground testing prior to certification for flight.
All in-person	COVID-19	Infected	2C	Team members will	3D	All meetings will	UNT policy



activities	Exposure/ Contracting the virus	persons spreading viruses to others in close proximity.		have ready access to masks and hand sanitizer, as well as handwashing sinks. Sick team members will not be permitted to come into the work areas or attend meetings in person.		be conducted in- person and online simultaneously to allow members to stay home and participate. TMs in the work areas will be limited to necessary personnel.	requires every person not vaccinated to get tested each month. All students are encouraged to wear masks while indoor. Team members will be asked to agree to limit risky activities for the good of the team.
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Table 2: ORM Assessment Worksheet [2]

5.5. NAR/TRA Procedures

NAR High Power Rocketry Safety Code [3]:

1. **Certification.** I will only fly high power rockets or possess high power rocket motors that are within the scope of my user certification and required licensing.
2. **Materials.** I will use only lightweight materials such as paper, wood, rubber, plastic, fiberglass, or when necessary ductile metal, for the construction of my rocket.
3. **Motors.** I will use only certified, commercially made rocket motors, and will not tamper with these motors or use them for any purposes except those recommended by the manufacturer. I will not allow smoking, open flames, nor heat sources within 25 feet of these motors.
4. **Ignition System.** I will launch my rockets with an electrical launch system, and with electrical motor igniters that are installed in the motor only after my rocket is at the launch pad or in a designated prepping area. My launch system will have a safety interlock that is in series with the launch switch that is not installed until my rocket is ready for launch and will use a launch switch that returns to the “off” position when released. The function of onboard energetics and firing circuits will be inhibited except when my rocket is in the launching position.
5. **Misfires.** If my rocket does not launch when I press the button of my electrical launch system, I will remove the launcher’s safety interlock or disconnect its battery and will wait 60 seconds after the last launch attempt before allowing anyone to approach the rocket.
6. **Launch Safety.** I will use a 5-second countdown before launch. I will ensure that a means is available to warn participants and spectators in the event of a problem. I will ensure that no person is closer to the launch pad than allowed by the accompanying Minimum Distance Table. When arming onboard energetics and firing circuits I will ensure that no person is at the pad except safety personnel and those required for arming and disarming operations. I will check the stability of my rocket before flight and will not fly it if it cannot be determined to be stable. When conducting a simultaneous launch of more than one high power rocket I will observe the additional requirements of NFPA 1127.
7. **Launcher.** I will launch my rocket from a stable device that provides rigid guidance until the rocket has attained a speed that ensures a stable flight, and that is pointed to within 20 degrees of vertical. If the wind speed exceeds 5 miles per hour, I will use a launcher length that permits the rocket to attain a safe velocity before separation from the launcher. I will use a blast



deflector to prevent the motor's exhaust from hitting the ground. I will ensure that dry grass is cleared around each launch pad in accordance with the accompanying Minimum Distance table and will increase this distance by a factor of 1.5 and clear that area of all combustible material if the rocket motor being launched uses titanium sponge in the propellant.

8. **Size.** My rocket will not contain any combination of motors that total more than 40,960 N-sec (9208 pound-seconds) of total impulse. My rocket will not weigh more at liftoff than one-third of the certified average thrust of the high-power rocket motor(s) intended to be ignited at launch.
9. **Flight Safety.** I will not launch my rocket at targets, into clouds, near airplanes, nor on trajectories that take it directly over the heads of spectators or beyond the boundaries of the launch site and will not put any flammable or explosive payload in my rocket. I will not launch my rockets if wind speeds exceed 20 miles per hour. I will comply with Federal Aviation Administration airspace regulations when flying and will ensure that my rocket will not exceed any applicable altitude limit in effect at that launch site.
10. **Launch Site.** I will launch my rocket outdoors, in an open area where trees, power lines, occupied buildings, and persons not involved in the launch do not present a hazard, and that is at least as large on its smallest dimension as one-half of the maximum altitude to which rockets are allowed to be flown at that site or 1500 feet, whichever is greater, or 1000 feet for rockets with a combined total impulse of less than 160 N-sec, a total liftoff weight of less than 1500 grams, and a maximum expected altitude of less than 610 meters (2000 feet).
11. **Launcher Location.** My launcher will be 1500 feet from any occupied building or from any public highway on which traffic flow exceeds 10 vehicles per hour, not including traffic flow related to the launch. It will also be no closer than the **appropriate Minimum Personnel Distance from the accompanying table from any boundary of the launch site.**
12. **Recovery System.** I will use a recovery system such as a parachute in my rocket so that all parts of my rocket return safely and undamaged and can be flown again, and I will use only flame-resistant or fireproof recovery system wadding in my rocket.
13. **Recovery Safety.** I will not attempt to recover my rocket from power lines, tall trees, or other dangerous places, fly it under conditions where it is likely to recover in spectator areas or outside the launch site, nor attempt to catch it as it approaches the ground.



MINIMUM DISTANCE TABLE

Installed Total Impulse (Newton-Seconds)	Equivalent High Power Motor Type	Minimum Diameter of Cleared Area (ft.)	Minimum Personnel Distance (ft.)	Minimum Personnel Distance (Complex Rocket) (ft.)
0 — 320.00	H or smaller	50	100	200
320.01 — 640.00	I	50	100	200
640.01 — 1,280.00	J	50	100	200
1,280.01 — 2,560.00	K	75	200	300
2,560.01 — 5,120.00	L	100	300	500
5,120.01 — 10,240.00	M	125	500	1000
10,240.01 — 20,480.00	N	125	1000	1500
20,480.01 — 40,960.00	O	125	1500	2000

5.6. Workspace Safety Brief

- All workspaces on the UNT campus have specific safety requirements and hazard training requirements associated with them. Any person not trained for any workspace will not be able to perform activities within that area or with the associated equipment until that training is completed.
- Appropriate personal protective equipment (PPE) will be always worn while working in any of the shop/lab spaces with hazardous machines or processes. These PPE items will be stored in each workspace and sourced/provided to each person by the team.
- Whenever working with paints, epoxy, and sanding, the appropriate level of respiratory protection will be worn.
- Disposable gloves, hearing protection, masks, and respirators will be provided whenever necessary.
- Closed toed shoes must be worn whenever working in the Aerospace workspace.
- Power tools will not be used unless there are two people present who are trained to use the tools safely.
- Report all incidents and near misses to the safety officer.
- Testing involving ejection charges will only be conducted in open spaces and under the supervision of the team's NAR/TRA mentor.
- Fire Extinguishers and emergency eyewash stations are placed throughout the J-wing, and it is everyone's responsibility to become familiar with their locations.
- A roster will be kept by the safety officer showing which safety training requirement each member has completed.
- Clean up all spills immediately.
- Keep the shop area clean and free of trash and tripping hazards.
- Backpacks and personal items will only be stored in designated areas.
- Whenever you are finished with any tools, return them to the toolbox.



- When you finish with power tools, ensure that they are turned off and unplugged.
- If a tool is broken, report it to the safety officer and the Robotics inventory manager so that it can be replaced.
- Rags, towels, and any chemical residue will be disposed of properly. Do not throw chemical-soaked towels in the garbage or pour excess into the drain. All epoxy/resin waste will be allowed to cure in the open and cooled before disposing of it.

5.7. NAR/TRA Mentor Actions

Support

- Mentoring team members who wish to certify level 1.
- Providing guidance and feedback during the project.
- Assist in planning for STEM outreach.

Design and Configuration

- Review preliminary and adjusted plans to include structural designs, motor type and specifications, recovery system design, payload and component designs, subscale vehicle design.

Safety

- Review all checklists to ensure team and vehicle flight readiness and safety.
- Review safety reports and updates.

Manufacturing

- Review and advise on manufacturing procedures and techniques.
- Review materials and plans for use.

Launches

- Supervise all launches.
- Purchase and handling of all level 2 motors, to include purchase, transportation, storage and preparation.
- Preparation of other energetic materials, i.e., black powder ejection charges.

5.8. Safety Briefing

All members of the team will be briefed on hazard recognition and accident avoidance through mandatory safety orientation sessions conducted aerospace division workspace, and through safety briefings conducted in the EMF on campus. A separate pre-launch meeting will be held prior to each test launch to brief team members on the regulations and safety codes that must be followed at the launch site.

6. Technical Design

6.1. General Vehicle Requirements [1]

1. The launch vehicle and payload must reach an apogee altitude of 4,000-6,000 feet above ground level (AGL). Target altitude must be declared at the PDR milestone and will be used for scoring.
2. The vehicle must carry at least two commercially available altimeters that are specifically designed for initiating rocketry recovery events.



3. The launch vehicle must be designed to be recoverable and reusable. It will not have more than (4) independent sections.
4. The launch vehicle must be designed to be made flight ready at the launch site within 2 hours of the FAA waiver opening.
5. The launch vehicle must be capable of remaining flight ready for 2 hours at minimum without losing functionality.
6. “All teams will successfully launch and recover a subscale model of their rocket prior to CDR.” [1] This can occur at any time from proposal award to CDR deadline. The requirements for subscale flight are listed in the 2023 NASA Student Launch Handbook section 2.18.
7. Full scale demonstration flights are required, and their stipulations are outlined in section 2.19 of the NASA SL Handbook.
8. The team’s name and launch day contact information will be in or on each separate section of the rocket.
9. Any Lithium polymer batteries must be distinguishable from other hardware and marked as a fire hazard.
10. Each team shall use commercially available e-matches or igniters. Hand-dipped igniters shall not be permitted.
11. Each team will use a single, commercially available, ammonium perchlorate composite propellant (COTS, APCP) motor which is certified by the National Association of Rocketry (NAR), Tripoli Rocketry Association (TRA), or the Canadian Association of Rocketry (CAR).
12. The launch vehicle will have a minimum static stability margin of 2.0 at rail exit, a minimum thrust to weight (T/W) ratio of 5.0:1.0 and will accelerate to a minimum of 52 fps at rail exit.
13. Lithium batteries will be clearly marked and labeled as a fire hazard.

6.1.1. General Vehicle Description

The launch vehicle will carry two independently redundant RAFCO Payload Systems (RAFCOPS), each of which will independently react to the RAFCO sequence broadcast by the NASA SLI team at the launch site. The airframe has been initially designed to carry two payloads capable of performing the same task, via independent deployment methods.

6.1.1.1. Forward Airframe/Nose Cone

The forward airframe consists of the nose cone, primary payload (PL1), forward body tube, primary and secondary drogue ejection charges, drogue parachute and recovery hardware. The drogue parachute and shock cord will join the forward airframe with the avionics bay/secondary payload system via U-bolts and quick links. The nose cone will also house an Apogee Components Simple GPS Tracker for vehicle recovery if line of site (LOS) is lost.

6.1.1.2. Avionics Bay

The avionics bay will couple the forward and aft airframes. It will be joined to both airframe sections with sheer pins that will break at each recovery event. The avionics bay will hold both a Missile Works RRC3 and Altus Metrum EasyMini redundant altimeters, battery, and dual-deployment electronics. The forward portion of this coupler will be divided and electrically shielded from the aft section, which will hold the redundant deployment electronics. In the forward portion of the AV-Bay there will be a second payload system (PL2) to ensure mission success.



6.1.1.3. Aft Airframe

The aft airframe joins the avionics bay to the fin can and houses the main parachute and recovery harness, as well as a second GPS tracker for vehicle recovery. The aft airframe will be connected to the avionics bay via the main parachute and shock cord, with quick links and U-bolts at each end. The aft airframe will be joined to the fin can assembly by removable nylon rivets.

6.1.1.4. Fin Can

The fin can is a removable design in the case that testing shows adjustments must be made, or damage is sustained during a test flight. It consists of a 20-inch x 75mm motor mount tube (MMT) and motor retainer, 3 plywood centering rings, 4 Nike-like fins built from a carbon fiber core, foam sandwich material, and fiberglass overlay. The fins will be through the wall (TTW) design and will be fixed to the centering rings, MMT, and Fin Can body tube with epoxy resin. By ensuring all components are affixed to each other, it forms a single mass to transfer the force from the L910 motor to the rest of the airframe during flight.

6.1.1.5. Rocket Model

An OpenRocket model shown in figure 4 shows the center of pressure (CP) as a red dot and the center of gravity (CG) as a blue dot. The design accounts for a 4-lb payload mass in the Forward Airframe and a 1.5-pound redundant payload system attached to the AV-Bay. The pre-launch vehicle weight including a full motor and payload tube is approximately 26.12 lbs. With the engine installed the CG is 76.174 inches from the tip of the nose cone and the CP is 87.918 inches from the tip of the nose cone. These measurements make our preliminary design stability 2.13 cal. The preliminary design has an overall length of 112 inches (9.33 feet) and a major diameter of 5.503 inches.

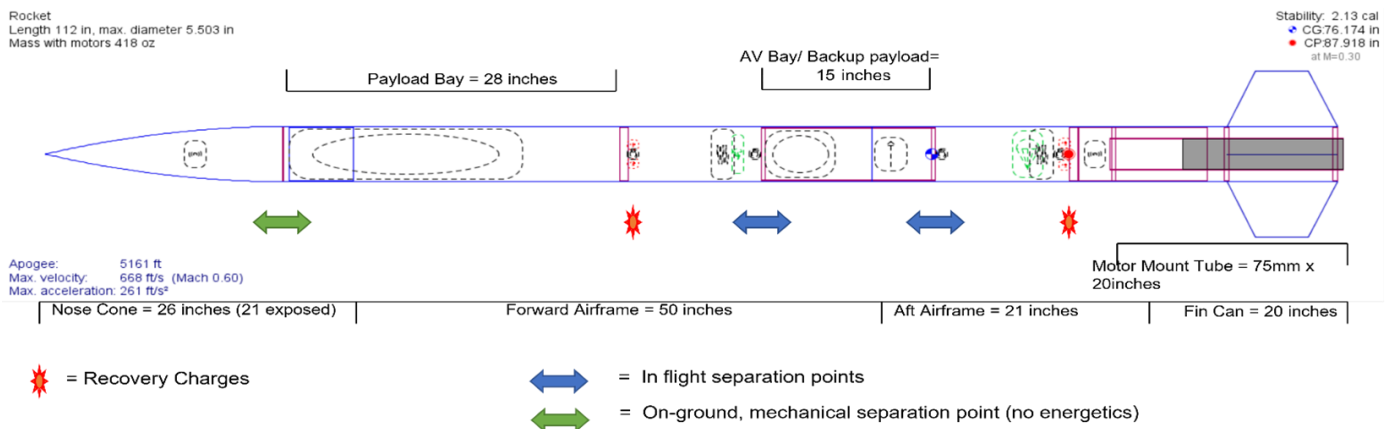


Figure 4: 2D Open Rocket model of the Vehicle



6.2. Preliminary Material Selection and Justification

Primary materials for construction of the launch vehicle will be:

1. woven fiberglass
2. carbon fiber
3. Blue Tube 2.0
4. plywood

6.2.1. Fiberglass

Mechanical Properties	Range	Units
Density	2.55-2.6	mg/m ³
Bulk Modulus	43-50	GPa
Compressive Strength	4k-5k	MPa
Ductility	0.026-0.028	
Elastic Limit	2750-2875	MPa
Hardness	3k-6k	MPa
Poisson's Ratio	0.21-0.23	
Shear Modulus	30-36	GPa
Tensile Strength	1950-2050	MPa
Young's Modulus	72-85	GPa

Table 3: Expected Mechanical Properties of Fiberglass [4]

6.2.2. Carbon Fiber

Mechanical Properties	Range	Units	Application Style
Young's Modulus	70	GPa	
Density	1.6	g/cc	
Tensile Strength	110	MPa	
Compressive Strength	110	MPa	
Shear Strength	260	MPa	In Plane
Poisson's Ratio	0.1		
Shear Modulus	5	GPa	

Table 4: Expected Mechanical Properties of Carbon Fiber [5]

6.2.3. Blue Tube 2.0

The vulcanized fiber which makes up Blue Tube 2.0 is strong enough to withstand the imparted setback forces on projectiles in the 155 mm Howitzer artillery system, which the manufacturer claims can reach up to 1000 G's. This makes it an ideal material for rocketry as it is a middle ground between the standard fiberglass/carbon fiber composites and the paper tubes. Through our own testing we have found that with repeated use Blue Tube can become brittle and is susceptible to thermal expansion, therefore we intend to reinforce the Blue Tube with a light (~4.0 oz.) E-glass to ensure reusability.



Mechanical Properties	Range	Units	Application Style
Tensile Strength	110	GPa	MD
Tensile Strength	62	GPa	CD
Compressive Strength	241	GPa	

Table 5: Mechanical Properties of Blue Tube 2.0 (Vulcanized Fiber)

6.2.4. Plywood

Laminated thin layers of wood makes for a lightweight and robust structure at a fraction of comparable material costs; therefore, we will be using plywood for most internal components like bulkheads and centering rings that we can laser cut or CNC to exact specifications. Structural plywood components can also be laminated with fiberglass or carbon fiber to enhance mechanical properties. The following table shows the expected mechanical properties of plywood without reinforcement/lamination.

Mechanical Properties	Range	Units	Application Style
Tensile Strength	26.7-34.5	MPa	Parallel to face
Modulus of Rupture	0.0483-0.0689	GPa	Parallel to face
Flexural Modulus	8.20-10.3	GPa	Parallel to face
Compressive Strength	31.0-41.4	MPa	Parallel to face
Shear Modulus	0.138-0.207	GPa	In Plane
Shear Modulus	0.586-0.758	GPa	Through thickness
Shear Strength	1.72-2.07	MPa	In plane
Shear Strength	5.52-6.89	MPa	Through thickness

Table 6: Mechanical Properties of Plywood [6]

6.3. Construction Methods

6.3.1. Airframe Fabrication

The primary method for constructing airframe sections and couplers will be using cut sections of Blue Tube 2.0. Light E-glass and epoxy resin will be used to enhance the Blue Tube and strengthen sections of the body tube, without blocking RAFCO transmission and GPS acquisition through major components. As a possible secondary method, we will also be testing a wet layup of fiberglass airframe tubes this year and conducting trade-analysis for cost, strength, and performance benefits. This technique uses a mandrel to simultaneously wrap woven fiberglass whilst impregnating it with epoxy. This method of fabrication allows us to control the wall thickness while ensuring that all airframe components have a uniform inside diameter, which is important for manufacturing matching coupler components, as well as planning for the sizing of internal payload tube components.

6.3.2. Fin Fabrication and Construction

We will construct the Nike-like fins using a carbon fiber laminate core, foam sandwich material for contouring and weight reduction, and several layers of fiberglass laminate on the surface to increase strength to withstand the forces of flight as well as the impact forces in the likely event that the fins strike the ground first on launch vehicle touchdown. Our intent is to use a fin-can design that can be replicated and replaced if needed or reused in the future.



6.3.3. Coupler, Avionics and Electronics Bay Fabrication

The couplers will be Blue Tube 2.0. The outside diameter tolerances are too close for these components to be over-wrapped with fiberglass, so if reinforcement is needed for structural integrity ribs and bulkheads can be incorporated internally for support. Additionally, sleds for carrying components within the avionics bay and nosecone may be constructed from either wood or fiberglass, or a combination of these.

6.4. Projected Altitude

Our preliminary design simulations show our maximum altitude above ground level will be approximately 5,161 feet. Our preliminary altitude declaration is 5,000 feet and will be adjusted by the PDR deadline.

6.5. Recovery System

6.5.1. General Recovery System Requirements [1]

1. The main parachute cannot be deployed lower than 500 feet
2. The apogee event cannot have longer than a 2 second delay
3. Motor ejection is not permissible
4. Every system/subsystem utilizing electronically initiated recovery events will be successfully ground-tested prior to being launched.
5. Maximum kinetic energy for any independent section of the vehicle will not exceed 75 ft·lbf at landing.
6. The recovery system will contain redundant COTS altimeters, each with its own dedicated, commercially available battery, capable of being turned on for at least 2 hours, each armed with a dedicated mechanical arming switch that can be locked in the “armed” position. These switches must be accessible from the exterior of the airframe when the rocket is in the launch configuration.
7. Recovery system electronics will be independent of payload electronics.
8. Removable shear pins will be used for main and drogue parachute compartments.
9. Recovery area is limited to 2,500 feet radius from the launch pad.
10. Descent time is limited to 90 seconds from Apogee to touchdown (Launch Vehicle).
11. An electronic GPS tracker will be installed in the launch vehicle, as well as the payload compartment and any untethered subsections, and each GPS tracker will actively transmit the position of each subsequent part to a ground receiver.
12. Recovery system electronics must be shielded from all other on-board electronic devices during the entire flight of the vehicle to ensure proper safe operation of recovery electronics.

6.5.2. Planned Recovery System Design

Dual deployment will be used for recovery in keeping with the general requirements in the Student Launch Handbook. Two dual deployment altimeter systems will be used, an Altus Metrum EasyMini altimeter, and a Missile Works RRC3 dual deployment altimeter. The EasyMini will be our primary altimeter because the RRC3 cannot be set to deploy exactly at apogee.

Dual altimeters with independent power sources will initiate an ejection charge as close to apogee as possible to deploy the drogue parachute without causing damage to the vehicle. At 700 feet AGL during



descent, the main parachute will be deployed to safely land the vehicle on the ground. The backup ejection charges will not exceed 2 seconds delay for each intended event.

After the launch vehicle has landed, servos will detach the parachutes from the avionics bay and the payload tube to prevent any wind from disturbing payload operations.

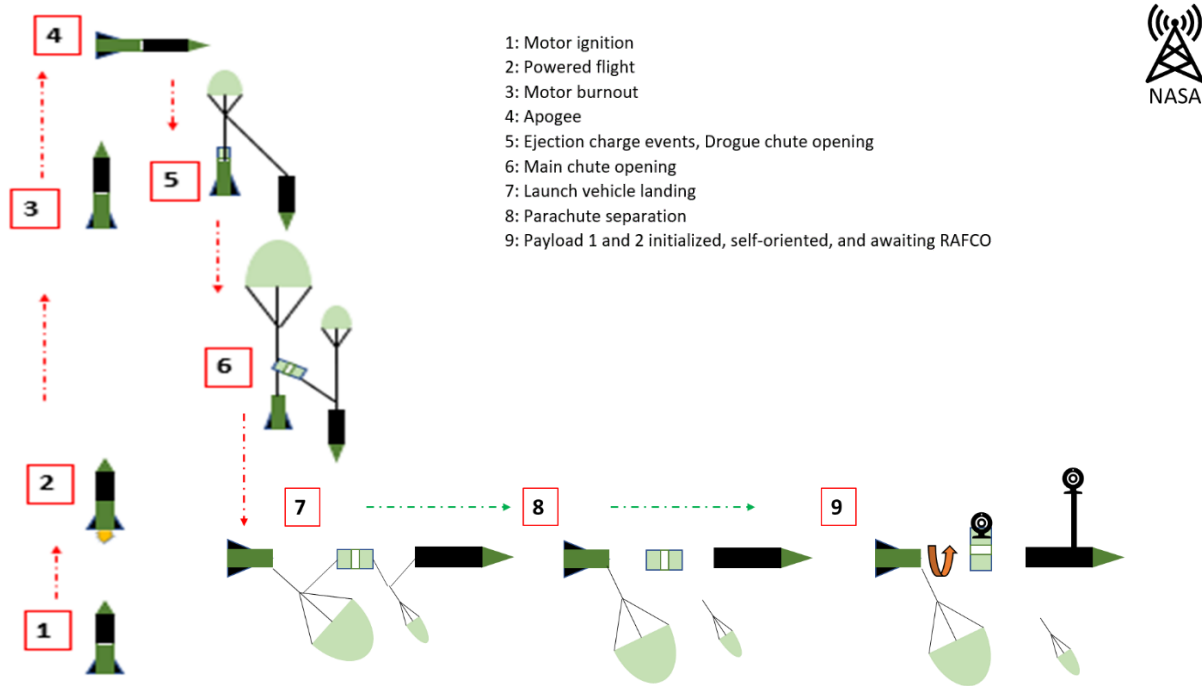


Figure 5: Recovery Diagram

6.6. Projected Motor Brand and Designation

The preliminary motor choice is the Cesaroni L910 75mm 2 grain motor. With an average/max thrust of 203.68 lb·f /235.6 lb·f [6] the launch vehicle's thrust to weight ratio is 7.79:1 and gives us an acceptable margin over the minimum T/W ratio of 5:1. The predicted apogee for this flight is 5000 ft AGL, and the launch vehicle velocity off rod is 87.3 ft/s on a 10 foot launch rail. Burn time for this motor is 3.16 seconds.

6.7. Payload

6.7.1. General Payload Requirements [1]

1. "Teams shall design a payload capable upon landing of autonomously receiving RF commands and performing a series of tasks with an onboard camera system. The method(s)/design(s) utilized to complete the payload mission shall be at the team's discretion and shall be permitted so long as the designs are deemed safe, obey FAA and legal requirements, and adhere to the intent of the challenge. [1]"
2. Launch vehicle shall contain an automated camera system capable of swiveling 360 degrees to take images of the entire surrounding area of the launch vehicle.



3. The camera shall have the ability to rotate about the z-axis (perpendicular to the ground).
4. Camera lens field of view must be between 100 and 180 degrees.
5. The camera must apply timestamps to all photos taken that can be visible in the photos submitted in the PLAR.
6. The payload must be able to quickly execute the string of actions, with no more than 30 seconds between photos taken.
7. The payload cannot initiate and begin accepting RAFCO until after the launch vehicle has landed on the surface.
8. The payload shall not be jettisoned.
9. It is not necessary to transmit the photographs back to the ground station, however they will be presented to NASA in the PLAR in the order they were taken.

6.7.2. Detailed Description of Projected Payload

6.7.2.1. Primary Payload (PL1)

The primary payload (PL1) will be a gyroscopic camera system carried in the forward payload tube. It will be on a folded mast within the payload tube. After landing, a servo will release the recovery harness to prevent wind on the ground from affecting the payload performance. Once the recovery harness is released, another servo will open the payload bay doors.

The payload bay doors account for two thirds of the payload surface area, so as the doors open, they will be used to orient and stabilize the payload tube in an upright position.

With the camera fully extended, the payload will begin listening for commands and execute the string of commands transmitted to it by NASA via RF.

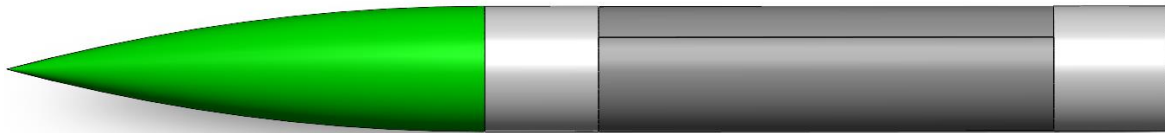


Figure 6: Payload Tube Configuration at landing

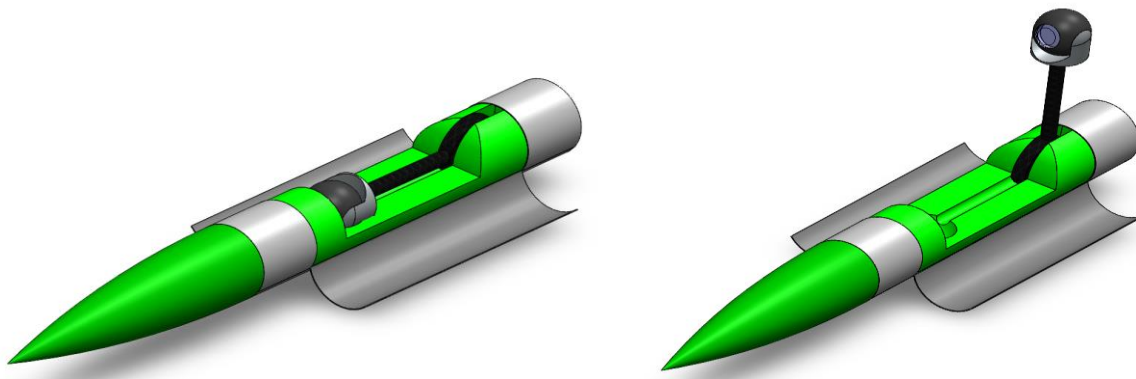


Figure 7: Payload Self-Orienting and Extending



6.7.2.2. Secondary Payload (PL2)

A redundant payload system (PL2) will be in the forward section of the coupler tube shared with the avionics bay. This payload will perform the same mission as the primary payload design. The difference is in the mechanical actions that will be taken by each payload to deploy the camera and stabilize/upright the tube. PL2 will be enclosed in a reinforced acrylic cylinder. The PL2/avionics bay coupler tube will use servos to release the recovery harness after landing to keep the payload from being disturbed by any winds catching the parachutes, just the same as PL1.

The avionics bay will structurally be the base of PL2 and will be connected to a group of spring-loaded arms. These arms will be released once the payload's gyro senses that it has come to rest, and their extension will upright PL2 so that the acrylic lens is oriented on top of the base, and the camera within can begin executing the commands transmitted by NASA and capturing photographs.

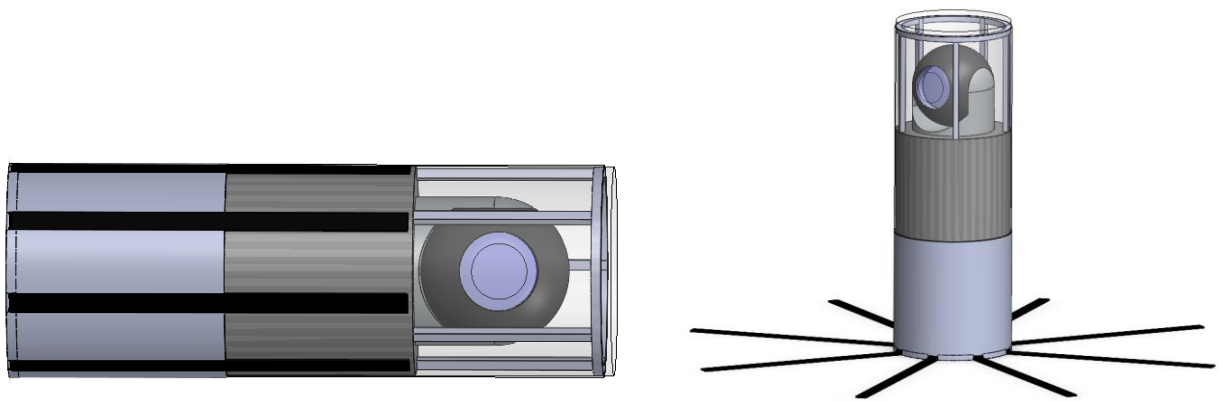


Figure 8: PL2 in Landed and Upright Configuration

6.7.3. Camera & APRS Receiver System

RAFCO commands transmitted via APRS will be received by a handheld radio, whose audio output will be plugged into a Raspberry Pi using a USB sound card. The Raspberry Pi will run software which can demodulate the audio signals into a string of text. These strings will be parsed by a python program on the Raspberry Pi which will then actuate the camera servos to orient it per the commands received and take and store the pictures as commanded.

6.7.3.1. RAFCO Transmission Receiving

We will use a COTS handheld receiver/transceiver, such as a Yaesu FT-65, to ensure we can reliably receive the transmissions. The radio will be partially disassembled to remove unnecessary weight and components. An audio jack will connect the speaker output to the soundcard on the Raspberry Pi, which will effectively transmit the raw AFSK audio to the Raspberry Pi for processing.

6.7.3.2. Parsing APRS & Commands

The raw AFSK data received by the soundcard will be piped into the program Dire Wolf (github.com/wb2osz/direwolf) which acts as an AX.25 packet modem. The resultant data from Dire Wolf

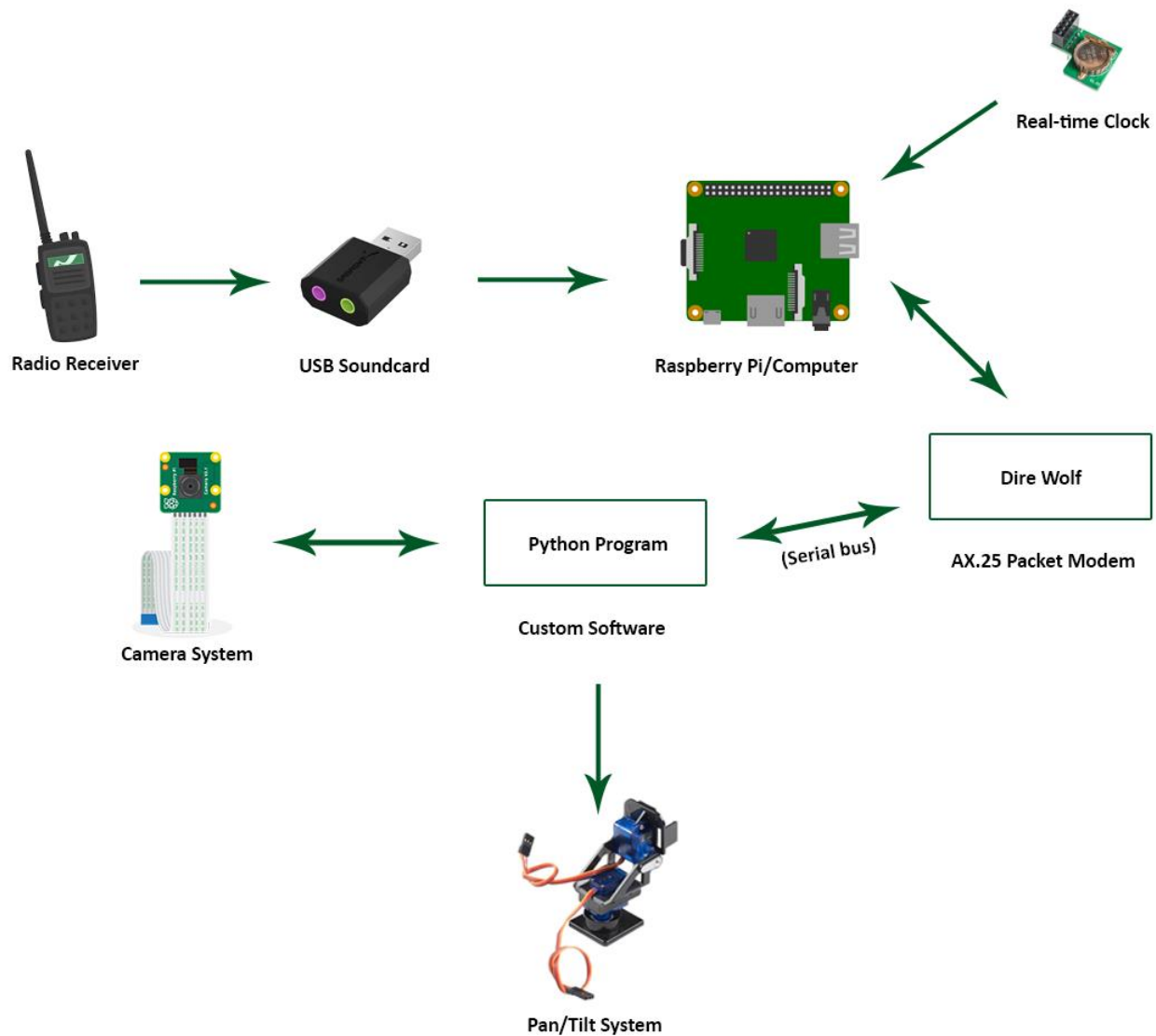


is made available via an internal serial channel (a pseudo terminal) and will be consumed by a custom written Python program.

The strings received by the Python program will be parsed to control the camera's orientation and capture pictures.

We will use a camera which is compatible with the Raspberry Pi CSI (Camera Serial Interface) port, which allows us to use the `picamera` python library, which also supports a suite of tools including the `annotate_text` property which easily adds timestamps to the images which are captures. A real-time clock module will be added to the Raspberry Pi to ensure the timestamps are accurate.

6.7.3.3. RAFCO & Camera System Diagram



6.8. Major Technical Challenges and Solutions

6.8.1. Airframe Fabrication

Last year our sub-scale vehicle was affected by repeated use. After three launches the body tube showed signs of failure likely due to a combination of heat and pressure from the ejection charges. To combat this issue, we intend to reinforce our body tubes with fiberglass to counteract the fatigue and thermal expansion issues of using Blue Tube repeatedly. This fix will add a small amount of mass to our rocket body but will make it more reusable than before. Another benefit of this method is that the final finish will be smoother than the original Blue Tube finish.

As a backup method, our team will be participating in a training event coordinated with another North Texas NASA SL team from the University of Texas at Arlington. Over the last two years we have developed a great, collaborative relationship between our two teams, and they have offered to teach us how they make their fiberglass tubes. We will be learning manufacturing skill from them, and if we can/find a need to, we will incorporate this skill into our design for the launch vehicle and its components.



Figure 6: University of Texas at Arlington and University of North Texas 2022 NASA USLI teams' joint subscale vehicle launch

6.8.2. Payload Design

The payload mission is the most challenging part of this entire project; finding a way to carry a camera system capable of performing the mission, and then finding a way for it to protrude out of/away from the launch vehicle in the correct orientation after landing, all while being structurally sound and resistant to the forces and vibrations of the flight. We have designed two separate payloads to perform the same mission to solve this problem.

Our designs for PL1 and PL2 draw inspiration from (1) the cargo bay doors on the NASA Space Shuttle, and (2) the British HB 876 area-denial landmine which is dropped from the air and uprights itself once on the ground. Using key concepts from these designs we hope to be able to accomplish the NASA SLI payload mission no matter what challenges the flight and terrain present to our vehicle and payloads.



7. Project Plan

7.1. Detailed Development Schedule

This is a tentative development schedule that will help to guide our team through the process for the next 8 months. The dates, including dates provided by NASA are subject to change but these are our current targets for the development cycle as we see it now. We will have to add test flight dates as we work with the local clubs and more milestones may be added as we progress.

Legend:

Payload	Manufacturing	Report Writing/Admin	Structural Analysis	Aerodynamics	Launch Day
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NASA Deadlines	Team Deadlines and Events
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August 2021

Request for Proposal Release	8/17/2022	Begin Payload Design	8/17/2022
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September 2021

Completed Proposal Due	9/19/2022	Initial Launch Vehicle Design Set (OpenRocket)	9/09/2022
		First Draft of Proposal Finished	9/09/2022
		First Draft of Proposal Revised	9/15/2022
		Establish component Criteria	9/15/2022
		DARS Sport Launch (Intro and familiarization event)	9/17/2022
		Final Proposal Revisions Done	9/18/2022
		Submit Final Draft of Proposal	9/18/2022

October 2021

Awarded Proposals Announced	10/04/2022	Payload Prototype 1 Started	10/04/2022
Kickoff and PDR Q&A	10/06/2022	Detailed Launch Vehicle Design (OpenRocket)	10/04/2022
PDR Due (8am CDT)	10/26/2022	First body Tube Layup	10/08/2022
		MATLAB structural limits calculations; SolidWorks FEA analysis	10/10/2022
		SolidWorks component designs finalized	10/10/2022
		Determine requirements for Parachute size, ejection charges, and	10/10/2022



		T2W ratio	
		Launch Opportunity	10/15/2022
		Cut body tubes, create bulkheads	10/16/2022
		PDR Rough Draft Complete	10/16/2022
		SolidWorks Sub-scale designs complete	10/16/2022
		Payload Prototype 1 Finished	10/19/2022
		Estimated altitude for full scale flight and launch rail exit velocity	10/19/2022
		PDR Revision Complete	10/24/2022
		Submit PDR Report	10/25/2022
		MATLAB and SolidWorks analysis of vehicle stresses and pressures	10/31/2022
		Finalize Payload design	10/31/2022

November 2021

PDR video teleconferences	11/01/2022-11/21/2022	Subscale Components manufacture	11/06/2022
		Begin testing subscale vehicle (ejection, recovery systems, electronics)	11/07/2022
		Complete Subscale Vehicle	11/13/2022
		Full/Subscale altitude calculations	11/17/2022
		Subscale Launch Opportunity #1	11/19/2022
		Fins	11/20/2022
		Fin Can	11/20/2022
		Updated Flight profiles (OpenRocket)	11/20/2022
		Rough Draft CDR Finished	11/24/2022
		Thanksgiving Break	11/24/2022 – 11/27/2022

December 2021

CDR Q&A	12/01/2022	Wood Fill Components	12/04/2022
		Begin Painting Full Scale Vehicle	12/04/2022
		CDR Revisions Complete	12/05/2022
		Final Subscale Launch Opportunity	12/17/2022
		Finals Week (No Required work)	12/12/2022 – 12/16/2022
		Winter Break	12/16/2022 – 1/16/2023



January 2022

Subscale Flight Deadline	1/09/2023	CDR Finished	1/05/2023
CDR Report and Presentation Due	1/09/2023	First Day of Spring Classes	1/16/2023
		CDR Submitted	1/08/2023
		Test Full Scale Components	TBD
CDR Video Teleconferences	1/17/2023- 2/07/2023	Make Fin Slots, finish electrical Components, Finalize all manufacturing	TBD
		Finish Full Scale Payload	TBD
		Full Scale Test Launch	1/21/2023

February 2022

FRR Q&A	2/09/2023	Rough Draft FRR complete	TBD
		Revisions for FRR complete	TBD
		Full Scale w/Payload Test Launch	2/18/2023

March 2022

Vehicle Demonstration Flight Deadline	3/06/2023	Submit FRR Report	3/05/2023
FRR Report and Presentation Due	3/06/2023	Submit FRR Addendum	As Needed
FRR Video Teleconferences	3/13/2023- 3/31/2023	Build Vehicle containers for Travel	3/29/2023
		Pack Supplies for Travel	3/29/2023
		Final Payload Flight Opportunity	3/18/2023

April 2022

Payload demonstration flight deadline	4/03/2023	PLAR completed	4/29/2023
Vehicle Demonstration re-flight deadline	4/03/2023	PLAR Submitted	4/30/2023
FRR Addendum Due	4/03/2023	Internal After-Action Report completed and archived	4/30/2023
Launch window for teams not traveling to Huntsville opens	4/01/2023		
Launch Week Q&A	4/06/2023		
Travel to Huntsville, AL	4/12/2023		
Optional LRR (Attending)	4/12/2023		
Official Launch Week Kickoff	4/13/2023		
Launch Week Activities	4/14/2023		
Launch Day	4/15/2023		



Backup Launch Day	4/16/2023		
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May 2022

PLAR Due	5/01/2023	Final Exams	5/07/2022 – 5/13/2022
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7.2. Project budget

Our project budget will include shop and safety materials, build and testing materials, and travel. This budget is mostly covered by last year's funding efforts, and any deficit will be made up from this year's sponsors. Supplies acquired from last year's competition that are reusable have significantly lowered our costs. The total budget at this point is between \$10,324.19 and \$11,524.19.

7.2.1. Full Scale and Shop Budget

The full-scale vehicle and payload total will be roughly \$3,868.45 to \$5,068.45 and includes \$200 to cover unforeseen expenses. The variation in cost accounts for differences in manufacturing processes, as well as the possibility of purchasing COTS electronics and photographic equipment to include in our payload.

Part	Note	Price	U/I	QTY	Total
Cesaroni L910 CS	2Grain	\$165.19	each	3	\$495.57
5.5" BT airframe (72" length)		\$105.00	each	4	\$420.00
5.5" ogive Nosecone		\$65.95	each	2	\$131.90
5.5" BT Coupler (48" length)		\$61.62	each	1	\$61.62
5.5" plywood Bulkhead		\$9.00	each	4	\$36.00
5.5"/75mm centering rings		\$7.50	each	8	\$60.00
75mm motor retainer		\$51.00	each	2	\$102.00
60" parachute		\$169.58	each	2	\$339.16
18" parachute		\$62.89	each	2	\$125.78
Shock Cord (28ft each)		\$33.14	each	4	\$132.56
U-bolts		\$8.23	each	12	\$98.76
Quick Links (1/4")		\$2.41	each	12	\$28.92
Rail Buttons (1515)		\$4.00	pair	2	\$8.00
Epoxy/Resin (RocketPoxy and West Systems)					\$200.00
Paint					\$75.00
Electronics					\$300.00 - \$500.00
Payload camera systems					\$400.00 - \$1,200.00
Composites					\$200.00 - \$400.00



GPS Tracker		\$453.18	Each	1	\$453.18
Other Expenses					\$200.00
				TOTAL	\$3868.45 - \$5068.45

7.2.2. Sub-Scale Production and Testing budget

We plan to build our subscale airframe from Blue Tube as well and have overestimated the motor cost for testing this vehicle. We will likely fly our subscale vehicle on a low impulse H or I motor so that we can match the Mach number anticipated for our full-scale flight, and to keep our subscale vehicle in a flight regime where we can record relevant data. The subscale budget total is currently \$655.74.

Item	Price	Quantity	U/I	Subtotal
Motor	\$75	3	each	\$225.00
Drogue Parachute	\$40.50	2	each	\$81.00
Recovery Hardware				\$30.00
Nose Cone	\$23.95	2	each	\$47.90
AV-Bay Coupler	\$40.00	2	each	\$80.00
3" x 48" airframe	\$30.96	4	each	\$123.84
3" coupler	\$9.00	2	each	\$18.00
Motor Retainer (38mm)	\$25.00	2	each	\$50.00
			Total	\$655.74

7.2.3. Team Travel Budget

The team plans to use University vehicles to drive to the launch with all our equipment and roughly 12-15 team members. Our travel budget will be requested through the university for our vehicles and per diem, and hotels and any other costs will be covered through sponsor finding.

Item	Cost
Transportation	\$800.00
Hotel	\$5,000.00
Total	\$5,800.00

7.3. Funding Plan

We are working with the university to find sponsors for the team and retain our previous sponsors. We intend to offer sponsors name and logo space on our team shirts, table-top presentation banner, social media page, and on the competition rocket body. We hope to appeal to industry leaders who have vested interest in the manufacturing practices and engineering that goes into design and development of a composite airframe containing a semi-autonomous robotic payload, as well as educating and mentoring students. We still have the support of last year's title sponsor, and we have received sponsorship promises from other companies for this year's competition season as well.



The university has also sponsored this project by allowing a senior design team to focus on this project as their capstone design project. This commitment is accompanied by \$1,000-\$1,500 sponsorship for materials from the university. With last year's funding surplus, and this year's anticipated funding, we are already fully funded for everything except travel for the entire season.

7.4. Sustainability Plan

We have established a mentoring system where new members can learn the engineering process involved in designing, building, and flying a project in the NASA SL competition directly from design leaders. We also have several active alumni who have been helping us financially with donations, equipment and connections with sponsors.

7.4.1. Establishing and Maintaining Partnerships

Our university has established relationships with local industry and community partners to facilitate achievement in STEM at the College of Engineering.

We are currently engaging potential sponsors and plan to use established relationships and UNT alumni to maintain/create partnerships with local companies going into the future. Our current sponsors, beginning with our title sponsor are RESPEC, Mouser Electronics, O'Reilly Auto Parts, University of North Texas, and UNT Robotics. We are still seeking more sponsorships from local aerospace companies.

7.4.1.1. Major Sponsorships and Partnerships



7.4.2. Engaging Successive Classes of Students in Rocketry

UNT Robotics actively recruits at introductory level courses in STEM majors at our university. The first two weeks of each semester our officers present to all introductory Mechanical, Electrical, and Computer Science/Engineering courses in person to generate interest in our project-based student organization and to inform the new underclassmen of our mission and purpose. Recruiting underclassmen is vital to our sustainability because with time comes experience and being able to work



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with students throughout their entire undergraduate careers allows our organization to successfully plan and complete exciting and advanced projects.

7.4.3. Team Member Recruitment

Team member recruitment is a continuing effort to ensure that our team has enough talent and ingenuity to succeed in the tasks set forth by NASA. Our current team is comprised mostly of Junior and Senior level engineering students who have demonstrated a drive to work on the project and potential for leadership. We are also recruiting from entry level engineering courses to bring in younger students who will learn to carry on the mission of our Aerospace Division.

7.4.4. Funding Sustainability

We have ongoing partnerships with last year's title sponsor RESPEC, and with local electronics and robotics equipment suppliers including Elegoo, ServoCity, O'Reilly Auto Parts, and Mouser, as well as Alumni who have made financial contributions to our organization. We intend to continue developing these relationships even as we look to create new partnerships.

7.4.5. STEM Engagement Activities

Our STEM engagement capabilities continue to grow as we build relationships and network within the community. We have continued to conduct outreach throughout the spring and summer, and by doing so we have made new relationships to continue our outreach. Our team is currently working with the university to participate in their outreach efforts, Tech Titans, The Enis STEM Academy, the local Scouts troop, and we are hoping to continue expanding on this network.

7.4.5.1. Evaluation Criteria

We will create an evaluation sheet for the parents/educators of the participants to evaluate our team members efficacy at presenting the material and engaging the participants. Older students will be given an opportunity to evaluate and give feedback as well. There will also be a verbal feedback opportunity with the participants to see if they enjoyed the event and learnt something new. Success criteria for our team at these events will be if the students have fun, learn something, and find new interests, and of course the event was conducted safely. As we head into this competition, the team is striving to not only enrich ourselves academically, but also engage K-12 students and instill into them a passion for discovery and the goal to pursue higher education.

The Dallas-Fort Worth area has more than 300,000 students pursuing K-12 education, many of which have already been inspired by NASA, SpaceX, Virgin Galactic, Blue Origin, Sierra Space, and many others to pursue STEM as a future career or get involved in STEM activities. The aim of the UNT Robotics Aerospace team is to lead an engaging experience in rocketry, space, and everything in-between that appeals to both students who are already engaged in STEM, and those who are new to it.

7.4.5.2. STEM at the Park

Date: October 2022 and March 2023

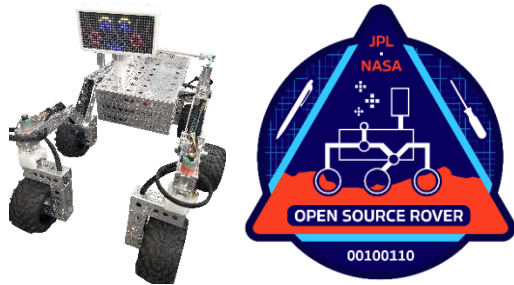


STEM at the Park is an event occurring each semester that offers adult, middle, and high school students the opportunity to experience engineering and computer science at the University of North Texas through hands-on projects and activities. The UNT Robotics Aerospace team will be hosting workshops to teach students about current space endeavors including the NASA SLS mission, and the mission of Perseverance and Ingenuity on Mars.

7.4.5.3. Mars Rover Project

Date: Ongoing

We have purchased the necessary materials to create the NASA JPL Open-Source Rover Project. This project is an ongoing effort that we have opened to all students and all majors at the university to foster creativity, interest, and innovation in robotics and has been ongoing for the past year. We host bi-weekly workshops to build segments of the rover and will have it completed by the end of the spring semester 2023.



7.4.5.4. Scouts STEM Engagement

Date: Late October

This event will be a partnership with Dallas Area Rocket Society, at which we engage the Scouts in building model rockets. Once built, they will have a competition to see who can land their rocket closest to a designated landing zone on “the Moon” which is a large tarpaulin placed at another spot on the launch field.

7.4.5.5. Tech Titans

Date: September 2022

This is an opportunity we have found through networking and hope to nurture a lasting relationship. It will not be submitted for the NASA SL STEM requirements because it is prior to the start of the season. Our first opportunity with them will be for our team lead to participate in a “speed mentoring” event at a local high school where the robotics program will divide the children into groups and the mentors will get about 15 minutes with each group to describe their path to engineering, give advice, and answer questions from the group.



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7.4.5.6. Lead to Achieve/Read Foundation

This summer we have been working on a commissioned video series for the Read Foundation. This is a local group that engages under-privileged children in STEM activities and provides them with a safe space and learning environment. The video series covers the applications and building of the Lego EV3 Mindstorms, and Lego Spike robotic kits and adds some fun anecdotes and learning opportunities while teaching the children to build, innovate, program with “scratch”, and think about general robotics applications from various angles. This effort will not be included in our outreach submissions but has led to more outreach opportunities for us in the future.

This fall we will be hosting a group of students at UNT from the Read Foundation to conduct STEM related workshops and inspire them by introducing the children to our projects.



Figure 9: Alumna Lauren Caves and Team Member Ibi Eni Filming Intro



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